

CBSE Class 12 Physics — Ray Optics and Optical Instruments

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark Questions (Objective/VSA)

- Q1.** A ray of light of wavelength **600 nm** propagates from air into a medium. If its wavelength in the medium becomes **400 nm**, the refractive index of the medium is: (a) 1.4 (b) 1.5 (c) 1.6 (d) 1.8. [PYQ 2023] Very High
- Q2.** An object is placed at **10 cm** in front of a concave mirror of radius of curvature **15 cm**. The position of the image is: (a) **15 cm** (b) **−30 cm** (c) **−25 cm** (d) **5 cm**. [PYQ 2022] Very High
- Q3. Assertion (A):** Propagation of light through an optical fibre is due to total internal reflection taking place at the core-cladding interface. **Reason (R):** Refractive index of the material of the cladding of the optical fibre is greater than that of the core. [PYQ 2023] Very High
- Q4. Assertion (A):** A convex mirror cannot form real images. **Reason (R):** Convex mirror converges the parallel rays that are incident on it. [PYQ 2020] High
- Q5.** A biconcave lens of power **P** vertically splits into two identical plano-concave parts. The power of each part will be: (a) **$2P$** (b) **$P/2$** (c) **P** (d) **$P/\sqrt{2}$** . [PYQ 2020] Very High
- Q6.** Two thin lenses are kept coaxially in contact with each other and the focal length of the combination is **80 cm**. If the focal length of one lens is **20 cm**, the focal length of the other would be: (a) **−26.7 cm** (b) **60 cm** (c) **80 cm** (d) **30 cm**. [PYQ 2023] High
- Q7.** A spherical air bubble is embedded in a piece of glass. For a ray of light passing through the bubble, it behaves like a: (a) converging lens (b) diverging lens (c) mirror (d) thin plane sheet of glass. [PYQ 2023] High
- Q8.** The magnification of an image by a convex lens is positive only when the object is placed: (a) at its focus **F** (b) between **F** and **$2F$** (c) at **$2F$** (d) between **F** and optical centre. [PYQ 2023] Very High
- Q9.** A convex lens of **20 cm** focal length forms a real image which is three times magnified. The distance of the object from the lens is: (a) **13.33 cm** (b) **14 cm** (c) **26.66 cm** (d) **25 cm**. [PYQ 2023] High
- Q10.** The critical angle for a pair of two media A and B of refractive indices **2.0** and **1.0** respectively is: (a) **0°** (b) **30°** (c) **45°** (d) **60°** . [PYQ 2022] Very High
- Q11.** The critical angle of pair of a medium and air is **30°** . The speed of light in the medium is: (a) **$1 \times 10^8 \text{ m/s}$** (b) **$1.5 \times 10^8 \text{ m/s}$** (c) **$2.2 \times 10^8 \text{ m/s}$** (d) **$2.8 \times 10^8 \text{ m/s}$** . [PYQ 2022] Very High
- Q12.** What is the focal length of a system of two lenses of power **8 D** and **−4 D** placed in contact? [Practice] Very High
- Q13.** For which of the following media, with respect to air, is the value of the critical angle minimum? (a) Crown glass (b) Flint glass (c) Water (d) Diamond. [PYQ 2022] High
- Q14.** A point source of light is placed at the bottom of a tank filled with water of refractive index **μ** to a depth **d** . What is the area of the surface of water through which light from the source can emerge? [PYQ 2022] High
- Q15.** When an object is placed between **f** and **$2f$** of a concave mirror, would the image formed be (i) real or virtual, and (ii) diminished or magnified? [PYQ 2015] Very High
- Q16.** What is the specific reason for the sparkling brilliance of a diamond? [Practice] High
- Q17.** A glass prism has internal angles of **45°** , **45°** and **90°** . The glass has a critical angle of **45°** . Does a ray incident normally on one of the shorter faces undergo total internal reflection at the hypotenuse? [PYQ 2025-26 SQP] Very High
- Q18.** A light ray entering a right-angled prism undergoes total internal reflection. What is the essential condition for the angle of incidence **i** on the reflecting face in terms of the critical angle **θ_c** ? [PYQ 2025-26 SQP] High
- Q19.** An astronomical telescope may be a refracting type or a reflecting type. Which of the two types produces an image of better optical quality (free from chromatic aberration)? [PYQ 2020] High

Q20. A symmetric biconvex lens of radius of curvature R is made of glass of refractive index 1.5 . What is its focal length in terms of R ? [Practice] Medium

2-Mark Questions

Q21. Define the term 'Total Internal Reflection'. State the two essential conditions under which this phenomenon takes place. [PYQ 2019] Very High

Q22. Use the mirror equation to rigorously show that a convex mirror always produces a virtual image, regardless of the object's placement. [PYQ 2016] Very High

Q23. A biconvex lens made of a transparent material of refractive index 1.25 is immersed in a tank of water having refractive index 1.33 . Will the lens behave as a converging or a diverging lens? Give the mathematical reason. [PYQ 2022] Very High

Q24. State with the help of a ray diagram, the working principle of optical fibers. Write one important practical use of optical fibers in communication or medicine. [PYQ 2019] High

Q25. Calculate the exact distance of an object of height h from a concave mirror of radius of curvature 20 cm , so as to obtain a real image of magnification 2 . [PYQ 2016] Very High

Q26. The focal length of a convex lens made of glass of refractive index 1.5 is 20 cm in air. What will be its new focal length when placed in a medium of refractive index 1.25 ? [PYQ 2022] High

Q27. An equiconvex lens forms a two-times enlarged real image when an object is kept 16 cm from it. The lens is cut into two identical plano-convex lenses. Determine the focal length of the original lens. [PYQ 2022] Medium

Q28. How does the refractive index of a transparent medium depend on the wavelength of the incident light used? Velocity of light in glass is $2 \times 10^8\text{ m/s}$ and in air is $3 \times 10^8\text{ m/s}$. Calculate the value of the critical angle for glass. [PYQ 2015] High

Q29. Show mathematically that the least possible distance between a real object and its real image in a convex lens is $4f$, where f is the focal length of the lens. [PYQ 2023] High

Q30. Lens Q when placed in contact with a converging lens P of focal length 20 cm makes a combination that behaves as a converging lens system of focal length 30 cm . Identify the exact nature of lens Q and determine its focal length. [PYQ 2025-26 SQP] Very High

Q31. Two monochromatic rays of light '1' and '2' are incident normally on the face AB of an isosceles right-angled prism ABC. The refractive indices of the glass prism for the two rays are 1.35 and 1.45 respectively. Trace the path of these rays after entering the prism. [PYQ 2014] High

Q32. Draw a schematic diagram of a reflecting type telescope (Cassegrain). Write its two important advantages over a refracting type telescope. [PYQ 2014] High

3-Mark Questions

Q33. A ray of monochromatic light passes through an equilateral glass prism in such a way that the angle of incidence is equal to the angle of emergence and each of these angles is $3/4$ times the angle of the prism. Determine the angle of deviation and the refractive index of the glass prism. [PYQ 2023] Very High

Q34. Draw the ray diagram of an astronomical telescope showing image formation in the normal adjustment position. Write the expression for its magnifying power. [PYQ 2019] Very High

Q35. A ray of light is incident on a glass prism of refractive index μ and refracting angle A . If it just suffers total internal reflection at the other face, obtain a relation between the angle of incidence, angle of prism, and critical angle. [PYQ 2023] High

Q36. Define the magnifying power of a compound microscope when the final image is formed at infinity. Why must both the objective and the eyepiece of a compound microscope have a significantly short focal length? Explain. [PYQ 2017] Very High

Q37. Calculate the radius of curvature of an equi-concave lens of refractive index 1.5 , when it is kept in a medium of refractive index 1.4 , to have a combined power of -5 D . [PYQ 2019] Medium

- Q38.** A point object in air is placed symmetrically at a distance of **60 cm** in front of a concave spherical surface of refractive index **1.5**. If the radius of curvature of the surface is **20 cm**, find the exact position of the image formed. [PYQ 2023] High
- Q39.** A concave mirror of focal length **12 cm** forms a three times magnified virtual image of an object. Find the distance of the object from the mirror. [PYQ 2023] Very High
- Q40.** An equilateral glass prism has a refractive index of **1.6** in air. Calculate the angle of minimum deviation of the prism, when it is kept in a medium of refractive index $\frac{4\sqrt{2}}{5}$. [PYQ 2019] High
- Q41.** A ray **PQ** of light is incident on the face **AB** of a glass prism and emerges out of the face **AC**. Trace the path of the ray. Plot a graph showing the variation of the angle of deviation as a function of the angle of incidence. State the condition under which δ is minimum. [PYQ 2015] Very High
- Q42.** The total magnification produced by a compound microscope is 20. The magnification produced by the eyepiece is 5. The microscope is focused on a certain object. The distance between the objective and eye-piece is observed to be **14 cm**. If the least distance of distinct vision is **20 cm**, calculate the focal length of the objective and the eye-piece. [PYQ 2014] High

5-Mark Questions (Long Answer/Case-Study)

- Q43.** (A) Draw a ray diagram for the formation of the image of a point object by a thin double convex lens having radii of curvature **R_1** and **R_2** . Hence derive the Lens Maker's formula. (B) A converging lens has a focal length of **10 cm** in air. It is made of a material of refractive index **1.6**. If it is completely immersed in a liquid of refractive index **1.3**, find its new focal length. [PYQ 2023] Very High
- Q44.** (A) Draw the ray diagram showing the refraction of a ray of light through a glass prism. Derive the expression for the refractive index μ of the material of the prism in terms of the angle of prism **A** and angle of minimum deviation δ_m . (B) Calculate the angle of emergence (**e**) of a ray of light incident normally on the face **AB** of an equilateral glass prism **ABC** of refractive index $\sqrt{3}$. [PYQ 2020] Very High
- Q45.** (A) With the help of a proper ray diagram, show how a concave mirror is used to obtain an erect and magnified image of an object. (B) Using the above ray diagram, systematically obtain the mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$ and the expression for linear magnification. [PYQ 2018] Very High
- Q46.** (A) Two thin lenses are placed coaxially in contact. Obtain the expression for the focal length of this combination **F** in terms of the focal lengths of the two individual lenses **f_1** and **f_2** . (B) A converging lens of refractive index **1.5** has a power of **10 D**. When it is completely immersed in a liquid, it mysteriously behaves as a diverging lens of focal length **50 cm**. Find the refractive index of this liquid. [PYQ 2020] Very High
- Q47.** (A) A point object 'O' is kept in a medium of refractive index **n_1** in front of a convex spherical surface of radius of curvature **R** which separates a second medium of refractive index **n_2** from the first one. Draw the ray diagram showing the image formation and deduce the relationship between the object distance **u** and the image distance **v** in terms of **n_1** , **n_2** , and **R** . (B) When the image formed above acts as a virtual object for a concave spherical surface separating the medium **n_2** back to **n_1** , draw the ray diagram and write the similar relation. Hence, obtain the Lens Maker's formula. [PYQ 2015] High
- Q48. Case Study: Optical Instruments.** A number of optical devices and instruments have been designed and developed utilizing the reflecting and refracting properties of mirrors, lenses, and prisms. (A) Why is the final image formed at infinity often considered most suitable for viewing through a telescope or microscope? Explain. (B) In modern microscopes, multi-component lenses are routinely used for both the objective and the eyepiece instead of single lenses. Why? (C) Write two significant points of functional difference between a compound microscope and an astronomical telescope. (D) What is the fundamental difference in the construction of the objective of a refracting telescope versus a reflecting telescope? [PYQ 2022] Very High
- Q49.** (A) A giant refracting telescope at an observatory has an objective lens of focal length **15 m**. If an eye-piece of focal length **1.0 cm** is used, what is the angular magnification of the telescope? (B) If this specific telescope is used to view the moon, what is the physical diameter of the image of the moon formed exclusively by the objective lens? (The diameter of the moon is $3.48 \times 10^6 \text{ m}$ and the radius of lunar orbit is $3.8 \times 10^8 \text{ m}$). [PYQ 2015] High

Q50. (A) Define the magnifying power of a compound microscope. (B) Draw a neatly labelled ray diagram of a compound microscope showing image formation at the least distance of distinct vision (D). (C) The sum of the focal lengths of the two lenses of a refracting telescope is **105 cm**. The focal length of one lens is **20** times that of the other. Determine the total magnification of the telescope when the final image is formed at infinity. [PYQ 2014] High

Answer Key

Q1: (b) 1.5. ($\mu_2/\mu_1 = \lambda_1/\lambda_2 = 600/400 = 1.5$). **Q2:** (b) -30 cm. ($f = R/2 = -7.5$ cm). $1/v = 1/f - 1/u = -1/7.5 + 1/10 = -1/30 \Rightarrow v = -30$ cm). **Q3:** (c) Assertion is true but Reason is false. (Cladding must have a lower refractive index than the core for TIR). **Q4:** (d) Both Assertion and Reason are false. (Convex mirrors *can* form real images if the object is virtual, and they *diverge* parallel rays). **Q5:** (b) $P/2$. (Splitting it vertically doubles the radius of curvature of the flat side to ∞ , halving the optical power of each half). **Q6:** (a) -26.7 cm. ($1/F = 1/f_1 + 1/f_2 \Rightarrow 1/80 = 1/20 + 1/f_2 \Rightarrow 1/f_2 = 1/80 - 4/80 = -3/80 \Rightarrow f_2 = -26.67$ cm). **Q7:** (b) diverging lens. (The bubble has a lower refractive index than the surrounding glass, causing it to diverge light). **Q8:** (d) between F and optical centre. (This produces a virtual, erect image, making m positive). **Q9:** (c) **26.66** cm. (Real image $\Rightarrow m = -3 \Rightarrow v = -3u$. $1/(-3u) - 1/u = 1/20 \Rightarrow -4/3u = 1/20 \Rightarrow 3u = -80 \Rightarrow u = -26.67$ cm). **Q10:** (b) 30° . ($\sin \theta_c = \mu_{rarer}/\mu_{denser} = 1.0/2.0 = 0.5 \Rightarrow \theta_c = 30^\circ$). **Q11:** (b) 1.5×10^8 m/s. ($\mu = 1/\sin(30^\circ) = 2$. $v = c/\mu = 3 \times 10^8/2 = 1.5 \times 10^8$ m/s). **Q12:** **25** cm. ($P_{eq} = P_1 + P_2 = 8 - 4 = 4$ D. $f = 1/P = 1/4$ m = **25** cm). **Q13:** (d) Diamond. (Diamond has the highest refractive index, so $\sin \theta_c = 1/\mu$ is minimum). **Q14:** Area = $\frac{\pi d^2}{\mu^2 - 1}$. (Derived from $r = d \tan \theta_c$ and $\sin \theta_c = 1/\mu$). **Q15:** (i) Real, (ii) Magnified. **Q16:** Diamond has a very high refractive index (2.42), giving it a very small critical angle ($\sim 24.4^\circ$). Light entering suffers multiple TIRs before emerging, causing it to sparkle. **Q17:** Yes. (The ray enters normal to the face, hits the hypotenuse at 45° . Since $i = 45^\circ \geq \theta_c$, it undergoes TIR). **Q18:** The angle of incidence i at the denser-to-rarer interface must be strictly greater than the critical angle θ_c ($i > \theta_c$). **Q19:** Reflecting type. (Mirrors do not suffer from chromatic aberration because reflection is independent of wavelength). **Q20:** $f = R$. ($\frac{1}{f} = (1.5 - 1) \left(\frac{1}{R} - \frac{1}{-R} \right) = 0.5 \times \frac{2}{R} = \frac{1}{R}$). **Q21:** TIR is the complete reflection of light back into the denser medium when it hits the interface at an angle greater than the critical angle. Conditions: (1) Light must travel from denser to rarer medium. (2) Angle of incidence $i > \theta_c$. **Q22:** $1/v = 1/f - 1/u$. For a convex mirror, $f > 0$. Real objects have $u < 0$. Thus $1/v = 1/f + 1/|u| > 0$. Since v is always positive, the image is always virtual and behind the mirror. **Q23:** Diverging. ($\frac{1}{f} = \left(\frac{\mu_{lens}}{\mu_{med}} - 1 \right) (\dots)$. Since $1.25 < 1.33$, the term $\left(\frac{1.25}{1.33} - 1 \right)$ is negative, making f negative). **Q24:** Principle: Total Internal Reflection. Use: Transmitting audio/video signals over long distances with negligible loss (or in endoscopy). **Q25:** $f = -10$ cm. Real image $\Rightarrow m = -2 = -v/u \Rightarrow v = 2u$. $1/2u + 1/u = 1/-10 \Rightarrow 3/2u = -1/10 \Rightarrow u = -15$ cm. **Q26:** $1/20 = (1.5 - 1)K \Rightarrow K = 1/10$. New: $1/f' = \left(\frac{1.5}{1.25} - 1 \right)K = 0.2 \times (1/10) = 1/50 \Rightarrow f' = 50$ cm. **Q27:** $m = -2 = v/u \Rightarrow v = 2 \times 16 = 32$ cm. $1/f = 1/32 - 1/(-16) = 3/32 \Rightarrow f = 32/3$ cm. **Q28:** Refractive index $\mu \propto 1/\lambda^2$ (Cauchy's formula). $\mu = v_{air}/v_{glass} = 3/2 = 1.5$. Critical angle $\theta_c = \sin^{-1}(1/\mu) = \sin^{-1}(2/3) \approx 41.8^\circ$. **Q29:** Let object distance be x . Image distance $v = D - x$. $1/(D - x) - 1/(-x) = 1/f \Rightarrow \frac{x+D-x}{x(D-x)} = 1/f \Rightarrow x^2 - Dx + Df = 0$. For real roots, discriminant $D^2 - 4Df \geq 0 \Rightarrow D \geq 4f$. **Q30:** $1/F = 1/f_P + 1/f_Q \Rightarrow 1/30 = 1/20 + 1/f_Q \Rightarrow 1/f_Q = 1/30 - 1/20 = -1/60$. $f_Q = -60$ cm. Lens Q is a diverging (concave) lens. **Q31:** Rays hit hypotenuse at 45° . Critical angle for Ray 1: $\sin \theta_{c1} = 1/1.35 = 0.74 \Rightarrow \theta_{c1} > 45^\circ$ (Ray 1 refracts out). Ray 2: $\sin \theta_{c2} = 1/1.45 = 0.68 \Rightarrow \theta_{c2} < 45^\circ$ (Ray 2 undergoes TIR). **Q32:** Advantages: (1) Free from chromatic aberration. (2) Easier to support a large mirror mechanically than a large heavy lens (resolving power is higher). **Q33:** $A = 60^\circ$. $i = e = \frac{3}{4} \times 60 = 45^\circ$. Deviation $\delta = i + e - A = 45 + 45 - 60 = 30^\circ$. Since $i = e$, δ is minimum. $\mu = \frac{\sin((60+30)/2)}{\sin(60/2)} = \frac{\sin 45^\circ}{\sin 30^\circ} = \frac{1/\sqrt{2}}{1/2} = \sqrt{2}$. **Q34:** Diagram showing objective, eyepiece, and rays forming image at infinity. $M = f_o/f_e$. **Q35:** Let incident angle be i , refracted angle r_1 . Hits second face at critical angle C . $r_1 + C = A \Rightarrow r_1 = A - C$. Using Snell's law at first face: $\sin i = \mu \sin r_1 = \mu \sin(A - C)$. **Q36:** Magnifying power $M = (L/f_o) \times (D/f_e)$. To achieve high magnification, both f_o and f_e must be as small as physically possible. **Q37:** $P = -5 = \frac{1}{f}$. $\frac{1}{f} = \left(\frac{1.5}{1.4} - 1 \right) \left(\frac{-2}{R} \right) \Rightarrow -5 = \left(\frac{0.1}{1.4} \right) \left(\frac{-2}{R} \right) \Rightarrow -5 = \frac{-2}{14R} \Rightarrow R = \frac{1}{35}$ m = $\frac{100}{35}$ cm ≈ 2.86 cm. **Q38:** $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$. Here $\mu_1 = 1$, $\mu_2 = 1.5$, $u = -60$ cm, $R = -20$ cm (concave to incoming light).

$\frac{1.5}{v} - \frac{1}{-60} = \frac{1.5-1}{-20} \Rightarrow \frac{1.5}{v} + \frac{1}{60} = \frac{-0.5}{20} = \frac{-1.5}{60} \Rightarrow \frac{1.5}{v} = \frac{-2.5}{60} \Rightarrow v = -36 \text{ cm. Q39: Virtual image} \Rightarrow m = +3.$
 $m = -v/u \Rightarrow v = -3u.$ Mirror formula:
 $1/v + 1/u = 1/f \Rightarrow -1/3u + 1/u = 1/(-12) \Rightarrow 2/3u = -1/12 \Rightarrow 3u = -24 \Rightarrow u = -8 \text{ cm. Q40:}$
 $\mu_{rel} = 1.6/(4\sqrt{2}/5) = 8/4\sqrt{2} = \sqrt{2}.$ For minimum deviation,
 $\mu_{rel} = \frac{\sin((A+\delta_m)/2)}{\sin(A/2)} \Rightarrow \sqrt{2} = \frac{\sin((60^\circ+\delta_m)/2)}{\sin(30^\circ)} \Rightarrow \sqrt{2} \times 0.5 = \sin((60^\circ+\delta_m)/2) \Rightarrow 1/\sqrt{2} = \sin 45^\circ \Rightarrow \delta_m = 30^\circ$
Q41: Draw $i - \delta$ parabola. Minimum deviation occurs when $i = e$, which means the refracted ray inside the prism is strictly parallel to the base. **Q42:** $M = M_o \times M_e \Rightarrow 20 = M_o \times 5 \Rightarrow M_o = 4.$ Eyepiece:
 $M_e = 1 + D/f_e \Rightarrow 5 = 1 + 20/f_e \Rightarrow f_e = 5 \text{ cm.}$ Eyepiece object distance:
 $1/(-20) - 1/u_e = 1/5 \Rightarrow 1/u_e = -1/4 \Rightarrow u_e = -4 \text{ cm.}$ Tube length
 $L = v_o + |u_e| \Rightarrow 14 = v_o + 4 \Rightarrow v_o = 10 \text{ cm.}$ Objective: $M_o = v_o/|u_o| \Rightarrow 4 = 10/|u_o| \Rightarrow u_o = -2.5 \text{ cm.}$
 $1/f_o = 1/10 - 1/(-2.5) = 1/10 + 4/10 = 5/10 \Rightarrow f_o = 2 \text{ cm. Q43:}$ (A) Derivation applying refraction formula to both surfaces and summing them. (B) $1/f_{air} = (\mu_g - 1)K \Rightarrow 1/10 = 0.6K \Rightarrow K = 1/6.$ In liquid:
 $1/f_l = (\mu_g/\mu_l - 1)K = (1.6/1.3 - 1)(1/6) = (0.3/1.3)(1/6) = 1/26 \Rightarrow f_l = 26 \text{ cm. Q44:}$ (A) Use
 $r_1 + r_2 = A, \delta = i + e - A.$ At $\delta_m, i = e \Rightarrow r_1 = r_2 = A/2, i = (A + \delta_m)/2.$ Apply Snell's law. (B) Normal incidence $\Rightarrow i = 0, r_1 = 0. r_2 = A = 60^\circ.$ At face AC, $\sin e = \sqrt{3} \sin 60^\circ = \sqrt{3}(\sqrt{3}/2) = 1.5,$ which is impossible ($\sin e \leq 1$). Thus, the ray undergoes Total Internal Reflection at face AC. **Q45:** (A) Diagram showing object between pole and focus, reflecting to form a virtual, erect, magnified image behind the mirror. (B) Use similar triangles $\triangle A'B'P \sim \triangle ABP$ to derive $m = h'/h = -v/u,$ and $\triangle A'B'F \sim \triangle MPF$ to derive the mirror formula. **Q46:** (A) Let objects and virtual images cascade. $1/v_1 - 1/u = 1/f_1$ and $1/v - 1/v_1 = 1/f_2.$ Summing gives
 $1/v - 1/u = 1/f_1 + 1/f_2 \Rightarrow 1/F = 1/f_1 + 1/f_2.$ (B) $P = 10 \text{ D} \Rightarrow f = 10 \text{ cm.}$
 $1/10 = (1.5 - 1)K \Rightarrow K = 1/5.$ In liquid,
 $1/-50 = (1.5/\mu_l - 1)(1/5) \Rightarrow -1/10 = 1.5/\mu_l - 1 \Rightarrow 1.5/\mu_l = 0.9 \Rightarrow \mu_l = 1.5/0.9 = 1.67. \text{ Q47:}$ (A) Derivation of $\frac{n_2}{v} - \frac{n_1}{u} = \frac{n_2 - n_1}{R}.$ (B) Apply same relation for second surface: $\frac{n_1}{v} - \frac{n_2}{v'} = \frac{n_1 - n_2}{-R_2}.$ Adding both yields Lens Maker's formula $\frac{1}{f} = (\frac{n_2}{n_1} - 1)(\frac{1}{R_1} - \frac{1}{R_2}).$ **Q48:** (A) Viewing an image at infinity relaxes the ciliary muscles of the eye, causing minimal eye strain. (B) Multi-component lenses minimize optical defects like spherical and chromatic aberration. (C) Microscope: $f_o < f_e,$ forms an enlarged image of a micro-object. Telescope: $f_o \gg f_e,$ forms an image of a distant large object to increase visual angle. (D) Refracting uses a large convex lens; reflecting uses a large concave parabolic mirror. **Q49:** (A) $M = f_o/f_e = 15 \text{ m}/0.01 \text{ m} = 1500.$ (B) Angle subtended by moon
 $\theta = D_{moon}/r_{orbit} = d_{image}/f_o.$
 $d_{image} = f_o \times (D_{moon}/r_{orbit}) = 15 \times (3.48 \times 10^6/3.8 \times 10^8) = 15 \times 0.00915 = 0.137 \text{ m} = 13.7 \text{ cm. Q50:}$ (A) Ratio of the angle subtended at the eye by the image to the angle subtended by the object at the unaided eye. (B) Standard diagram with final virtual image at distance $D.$ (C) $f_o + f_e = 105. f_o = 20f_e.$ Thus
 $21f_e = 105 \Rightarrow f_e = 5 \text{ cm, } f_o = 100 \text{ cm.}$ Magnification $M = f_o/f_e = 100/5 = 20.$